

ResNet-18 for EMNIST Classification

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I. INTRODUCTION

This report presents the implementation and evaluation of a ResNet-18 model for classifying handwritten digits (0-9) and uppercase letters (A-Z) from the EMNIST dataset. ResNet-18, a deep convolutional neural network with residual connections, was trained and evaluated to assess its classification accuracy.

II. DATASET PREPROCESSING

The EMNIST dataset 1 underwent several preprocessing steps to ensure optimal input representation. Images were first decoded and normalized, resized to 28x28 pixels, and scaled to the range [0,1]. To enhance generalization, random horizontal flipping and contrast adjustments were applied as part of data augmentation. Labels were one-hot encoded to match the 36-class classification task.

The dataset was split into 80% training and 20% validation, ensuring that data was shuffled before splitting. To optimize training efficiency, both datasets were batched (batch size = 64) and preloaded using prefetching.

III. MODEL ARCHITECTURE

ResNet-18, a deep convolutional neural network with residual connections, was chosen for its robust feature extraction capabilities and resilience to vanishing gradients.

The model consists of several key components designed to extract hierarchical features and facilitate gradient flow. The **initial convolutional layer** applies a **5x5 convolution with 64 filters and a stride of 2**, followed by batch normalization and ReLU activation for stable training. A **3x3 max pooling layer with stride 2** further downsamples the input, reducing computational complexity while preserving spatial features.

The core of the model consists of four residual blocks that progressively extract hierarchical features. **Block 1** comprises two **identity blocks with 64 filters**, which help maintain the feature dimensions while improving gradient flow through skip connections. **Block 2** introduces downsampling via a **convolutional block with 128 filters**, followed by an **identity block** to refine representations at this scale. **Block 3** continues this pattern with a **convolutional block using 256 filters** and an additional **identity block**, ensuring deeper feature extraction. Finally, **Block 4** further downsamples the feature map using a **convolutional block with 512 filters**, followed by an **identity block** to consolidate high-level representations.

To finalize classification, the model employs a **global average pooling layer**, reducing spatial dimensions and aggregating key features before passing them to a **fully connected (dense) layer with softmax activation**. **Skip connections**

throughout the network mitigate gradient vanishing and enhance information flow, ensuring an optimal balance between feature retention and computational efficiency.

IV. TRAINING AND EVALUATION

A. Training Setup

The model was trained for 20 epochs using the Adam optimizer with a learning rate of $1e-4$ and Categorical Cross-Entropy as the loss function.

To improve generalization and efficiency, several training callbacks were employed, including model checkpointing to save the best model based on validation accuracy, early stopping to monitor validation loss with a patience threshold of 5 epochs, and learning rate scheduling that reduced the learning rate by 0.5 if validation loss plateaued for 3 consecutive epochs to ensure stable convergence and mitigate the risk of overfitting.

B. Training Performance and Hyperparameter Tuning

The model started with an initial accuracy of 80.30% on training data and 90.50% on validation data in the first epoch. Over subsequent epochs, training accuracy steadily increased, reaching 95.35%, while validation accuracy improved to 93.22%.

Training loss followed a downward trend, reducing from 0.6324 in the first epoch to 0.1114 at the end of training. Validation loss decreased but stabilized around epoch 18, indicating effective learning and generalization.

Learning rate adjustments were triggered at epoch 9 (reducing from $1e-4$ to $5e-5$) and again at epoch 19 (further reducing to $2.5e-5$), preventing overfitting and maintaining performance stability.

V. RESULTS

- **Precision (0.9392):** The model correctly predicts 93.92% of the time, indicating minimal false positives.
- **Recall (0.9290):** The model captures 92.90% of actual instances, though some misclassifications remain.
- **F1-score (0.9329):** This metric balances precision and recall, reflecting overall classifier robustness.

The confusion matrix 3 reveals that the model performed well across most categories. However, minor misclassification errors occurred, particularly among similar-looking characters such as 'O' and '0' 4, and additional preprocessing techniques could help further improve performance.

VI. APPENDIX

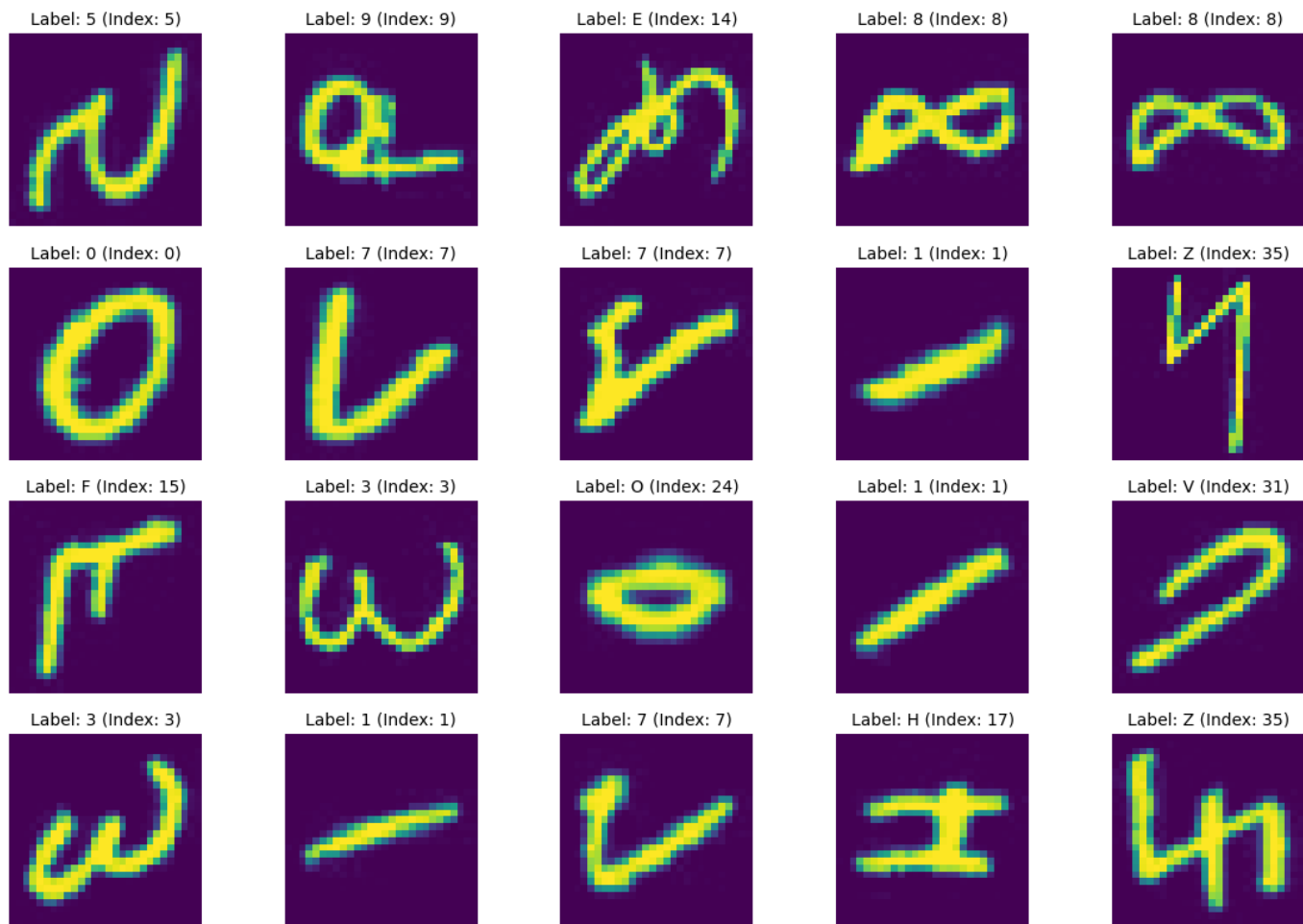


Fig. 1. Dataset Overview

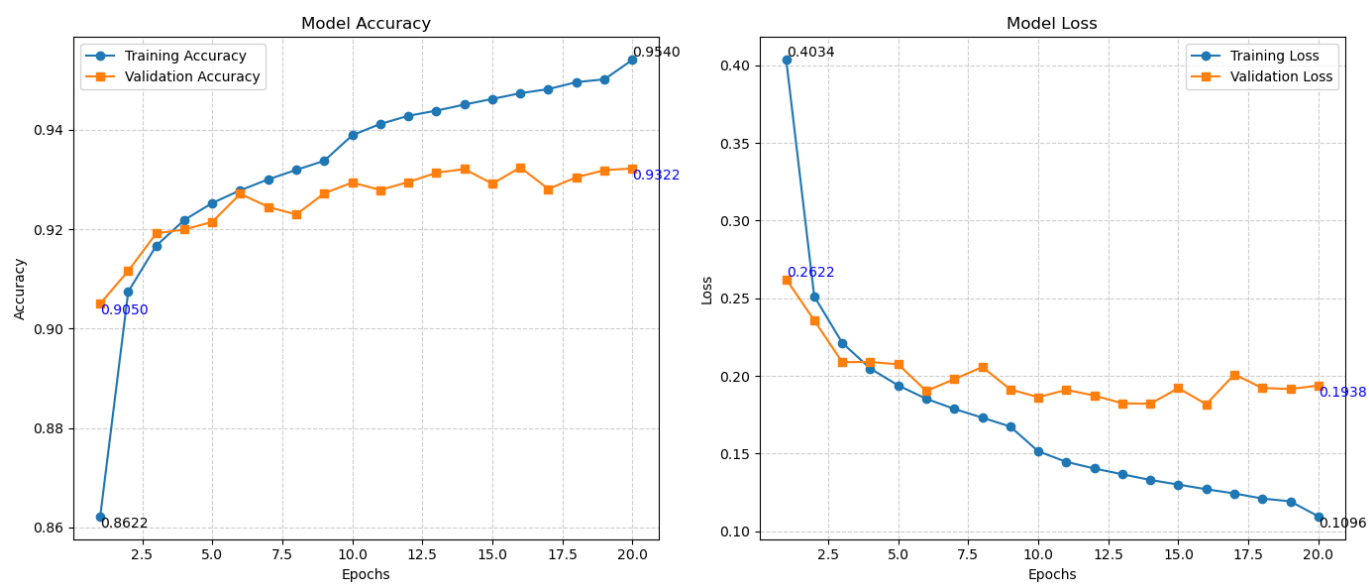


Fig. 2. Training and Validation Metrics

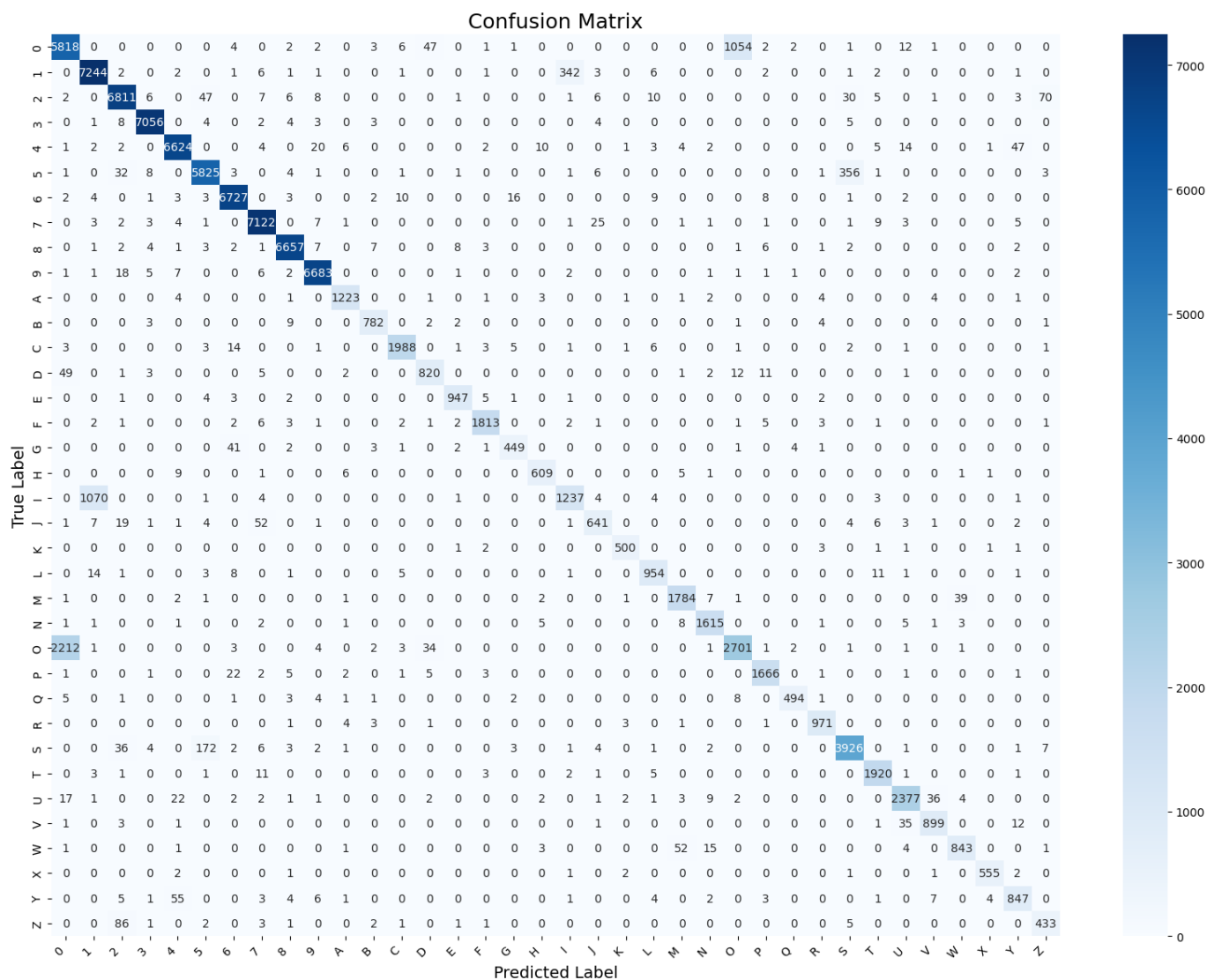


Fig. 3. Confusion Matrix Showing Classification Performance

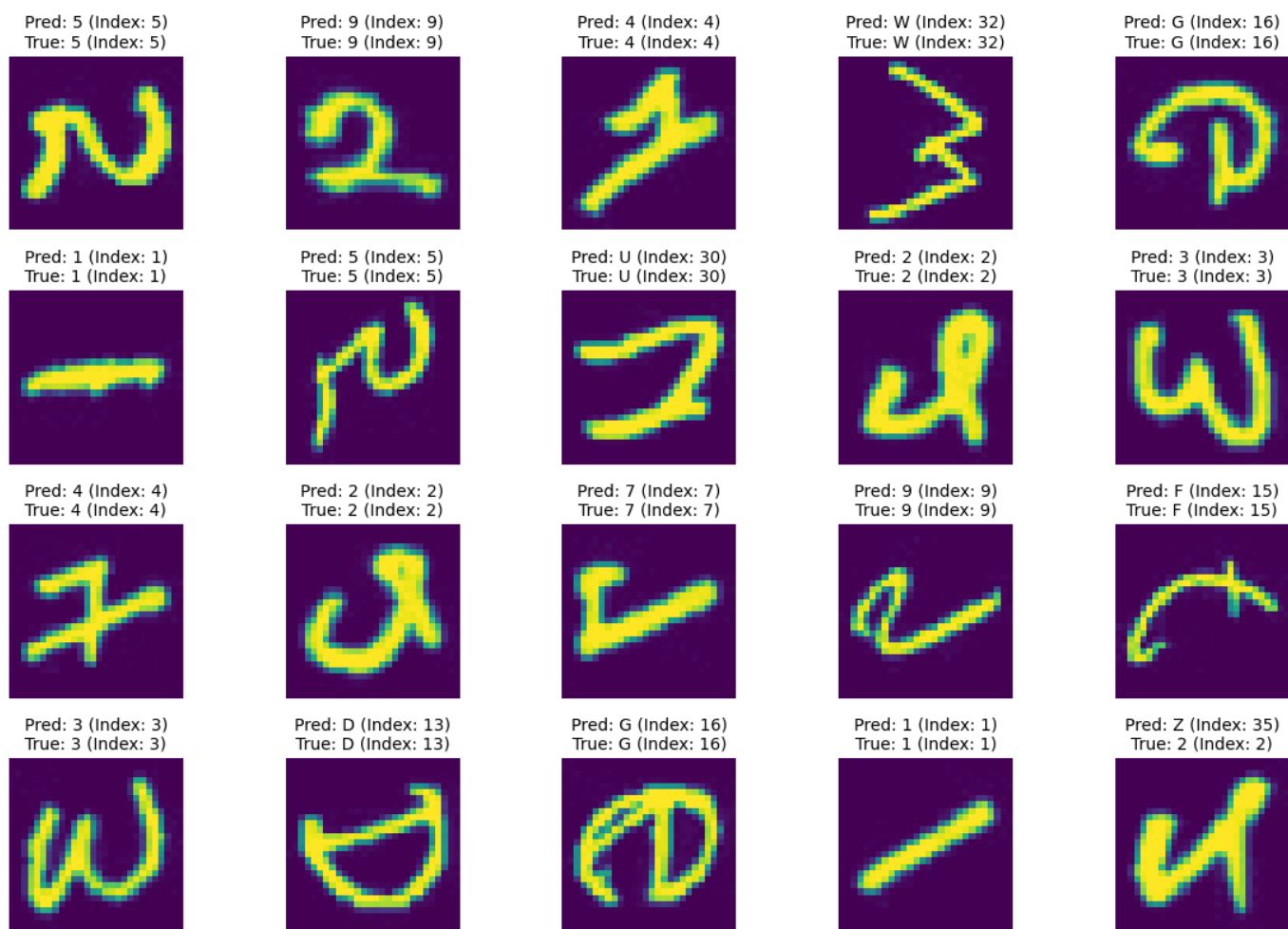


Fig. 4. Sample Predictions from the Test Set